Steppan Konoplev

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EDUCATION

2022-Present Math PhD at University of Delaware (UD)

- Advised by Mokshay Madiman
- Completed all requirements except thesis, working on research to begin thesis

2022-2023 Masters of Science in Mathematics at UD

- Finished in 1st year of PhD by acing qualifying exams and satisfying credits requirement

2019-2022 University of Maryland College Park (UMD) math major

- B.S. in Mathematics with High Honors, also CS minor
- Science, Discovery, and the Universe Scholars Program

PUBLICATIONS AND PREPRINTS

- Konoplev, S., Medel, J., & Russell, V. (2025). Non-Kahler C-Y 3-Folds Arising from Singular K3 Surfaces. *Note Di Matematica, 45*(2).
- Konoplev, S., Medel, J., & Russell, V. (2022). Cohomology of K3 surfaces. Technical report for AMRPU 2022
- Konoplev, S. (2019). On Alphatrion's Conjecture about Hamiltonian Paths in Hypercubes. arXiv. Retrieved from https://arxiv.org/abs/2002.02285
- Konoplev, S. (2019). Convergence of General Alternating Series. *CHS Math Journal*, 30–32.
    - drive.google.com/file/d/1dsfzjESEndIhoFnpD2aDSLmhU5w9qisb

PRESENTATIONS

- Talk Series, Operator Algebras Seminar, UD, November – December 2023, *Classification of von Neumann Algebras* (and prerequisite results)
- Talk, HGSS, UD, 20 September 2023, *Seriation of Samples of Graphons*
- Talk, GTA Philadelphia Conference, Temple University, 28 May 2023, *Calculating Cohomology of singular K3 surfaces*
- Invited Talk, Graduate Student Intercollegiate Mathematics Seminar, Lehigh University, 25 April 2023, *Probabilistic Method for solving Discrete Math Problems*
- Poster, UD Winter Research Symposium, University of Delaware, 3 March 2023, *Cohomology of Singular K3 surfaces*
- Talk, HGSS, UD, 20 February 2023, *Induction on the Real Numbers*
- Talk, HGSS, UD, 19 October 2022, *Probabilistic Method*

ACADEMIC COMPETITIONS AND AWARDS

- 7th in 2024 Intercollegiate Programming Competition Mid-Atlantic Site Regionals
- 1st place in 2023, 2024, 2025 George Mason University Calculus Olympiad
- UD 2023 Winter Research Symposium Winning Poster
- 1st place in 2022 New Jersey Undergraduate Math Competition
- UMD comprehensive math honors exam all time high score
- 2021 IMC (International Math Competition) first prize, highest ranked U.S. student, 32nd/589
- Top 50 in 2020 Putnam Math Competition (based on score of 68/120 and 39 for top 100)
- 2020 IMC first prize
- 1st place in 2019 Virginia Tech Regional Mathematics Competition
- Top 200 in 2019, 2021 Putnam Math Competition

- Lockheed Martin Challenge Box: Won Voyager Golden Record replica for solving challenging computer science problems

RESEARCH AND PROJECTS

May 2024-ongoing — Candidacy, Research, and Thesis with Mokshay Madiman

- Directions and project ideas planned for research work, thesis topic TBD
- Candidacy exam passed in December 2024
  - Covered *The Expected L_p Norm of Random Polynomials,* Borwein-Lockhart

January 2023-September 2023 — Graph Limits with Mahya Ghandehari

- Read chapters 7-11 of Lovasz's textbook *Large Networks and Graph Limits* in spring 2023 and presented proofs of multiple major results in informal graphons seminar
- Collaborated with Mahya and Jeannette Janssen to generalize *A Spectral Algorithm for Seriation* (1998) to graphons
  - Routine parts of generalization published by collaborators
  - Original contributions to be gathered and appear in a solo paper in 2025

February 2023-April 2023 — Projective geometry with Robert Coulter

- Worked on open problem: projective plane having linear planar ternary ring implies transitive elation or transitive homology group
- Positive answer implies significant simplification of Lenz-Barlotti classification
- Disproved claim by prominent projective geometers that Pickert's book *Projektive Ebenen* implies a proof, explained why problem still open and out of reach of coordinatization methods

May 2022-July 2022 — Florida International University REU

- Researched complex algebraic geometry with Gueo Grantcharov and Anna Fino
- Completed commutative algebra worksheets, learned basic material on K3 surfaces, and read papers on weighted projective spaces
- Extended Iano-Fletcher's paper *Working with Weighted Complete Intersections* by calculating the cohomology of all K3 surfaces with only A_n-type singularities

TEACHING AND RELATED JOBS

Aug 2022-Present — UD Math TA and Instructor

- Run discussion section, hold office hours every week, grade quizzes & exams
- Classes taught: finite mathematics (MATH 230), all levels of Calculus (MATH 241, 242, 243)
- No teaching in spring 2025, MATH 243 instructor in summer 2025, TA all other semesters

Aug 2021-May 2022 — Gossett Student Athletic Center Tutor

- Tutored student athletes in upper-level undergraduate math courses, including linear algebra and introduction to real analysis

Nov 2020-Feb 2022 — Office of Multi-ethnic Student Education Tutor

- Tutored OMSE students in any 100 or 200 level math or computer science course
- Managed GroupMe group for efficient sharing of information during tutoring sessions

Aug 2021-Dec 2021 — UMD MATH 410 grader

- Graded assignments for a real analysis class and gave feedback on proof writing

Jan 2021-May 2021 — UMD MATH 310 grader

- Graded assignments for an introduction to proofs class and gave feedback on proof writing

OTHER PROFESSIONAL ACTIVITIES

- Quora math writer (2017-21): Spent hundreds of hours answering math questions on Quora, with 263 answers and 600k views (including content collapsed by bots)
  - https://www.quora.com/profile/Steppan-Konoplev-1

- Student organizer at UD (2022-5): President and founder of Newark Card Games Club, social events organizer for Competitive Programming Club, Quizbowl Club tournament director, recruited dozens of players to Hen Zone table tennis
- Table tennis and pickleball player at UD
  - Awards include 1st in Spring 2026 UD pickleball club tournament, 1st in (Fall 2025 grad division, Summer 2025 mixed, Spring 2025 men's, Fall 2024 mixed) pickleball intramurals, 2024 UD table tennis intramural champion